

Saptaparno Chakraborty

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SKILLS

Languages:	C++, C, JavaScript/TypeScript
Frameworks/Libraries:	Node.js, Express, React, FTXUI
Databases:	MySQL, SQLite, MongoDB
Tools/Platforms:	Git, Linux, Vim/Neovim, CMake

PROJECTS

SeptCrawler | C++, Node.js | [Link](#) Apr' 26 - Present

- Designed a modular search engine for learning-focused retrieval, targeting 4+ content categories (documentation, references, tutorials, discussion forums) with a multi-stage ingestion and retrieval pipeline.
- Completed initial high-level and low-level system design, defining 8+ core components across C++ search engine and Node.js orchestration layers, including crawler, parser, indexing engine, ranking pipeline, and inverted index structures for efficient query processing.
- Defined system data flow and component interactions across 2 major pipelines (ingestion + query flow), focusing on extensible ranking strategies and intent-aware result organization, preparing the system for implementation.

WorkshopHub | Node.js, Express, MongoDB, React, JWT, bcrypt | [Link](#) Nov' 25 - Jun' 26

- Developed a full-stack workshop management platform supporting 3 user roles, to streamline 5+ end-to-end workflows of educational workshops, including registrations, attendance, materials, feedback, and certificates.
- Designed and implemented 20+ RESTful APIs with JWT authentication, bcrypt-based password security, and role-based access control (Admin, Instructor, Participant), along with 7 MongoDB schemas modeling the complete lifecycle.
- Engineered a scalable system with secure access control and modular architecture, gaining hands-on experience in backend design, API structuring, and full-stack integration.

Priorify | C++, STL, SQLite, FTXUI, CMake | [Link](#) Jul' 25 - Present

- Built a priority-based task manager supporting 6 core operations (add, update, delete, list, complete, clear) to efficiently organize tasks with deterministic ordering and fast local execution.
- Implemented core functionality in C++ using `std::priority_queue` with a custom comparator and a heap→vector→heap rebuild approach, structured across 4 core source modules (db, task_manager, tui, main) and transitioned from a manual Bash-based build pipeline to a modular CMake system with 3 external system dependencies (sqlite3, pthread, dl), and FetchContent-based dependency management for FTXUI.
- Strengthened understanding of data structures and persistence design through SQLite integration, handling persistent storage across sessions, and improved usability by evolving the interface from basic console I/O to a terminal UI using FTXUI.

TRAINING

CodeQuest: DSA Summer Bootcamp Jun' 25 - Jul' 25

Data Structures and Algorithms

- Focused on core and advanced Data Structures & Algorithms.
- Emphasized algorithmic optimization, problem-solving, and real-world software applications.
- Included daily hands-on coding, debugging, and complexity analysis sessions.

CERTIFICATES

Computational Theory: Language Principle & Finite Automata Theory Infosys Springboard	Aug' 25
Privacy and Security in Online Social Media NPTEL	Apr' 25
Fundamentals of Network Communication University of Colorado	Nov' 24
Introduction to Hardware and Operating Systems IBM	Sep' 24
The Bits and Bytes of Computer Networking Google	Sep' 24
Responsive Web Design freeCodeCamp	Nov' 23

EDUCATION

Lovely Professional University

Bachelor of Technology - Computer Science and Engineering; **CGPA: 7.87**

Punjab, India
Aug' 23 - Present